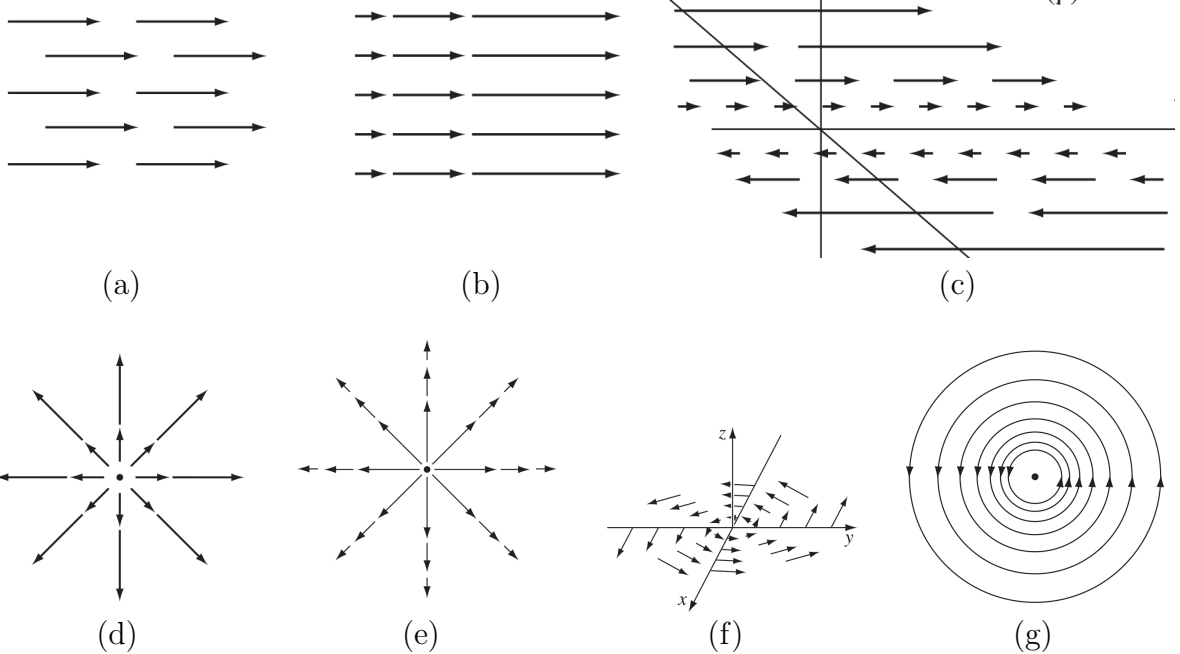


Relativity and Electromagnetism Exercise 1: 2026 Feb 11

Problem 1. Look at the following vector field configurations, and try to guess for each case whether the (i) divergence and (ii) curl is zero or non-zero. If the curl is non-zero, in which direction is it pointing?



Try to come up with expressions for each of these vector fields $\mathbf{v}(\mathbf{r})$ in a suitable coordinate system. Compute the divergence and the curl using these expressions and the standard formulae (available in Appendix C of the formula sheet). Do these answers agree with your initial guesses?

Note: For parts (d) and (e) you can assume $|\mathbf{v}(r)| \propto r$ and $|\mathbf{v}(r)| \propto 1/r^2$ respectively in spherical coordinates. For (f) and (g) you can assume $\mathbf{v}(\varrho) \propto \varrho$ and $\mathbf{v}(\varrho) \propto 1/\varrho$ respectively in cylindrical coordinates, with no z -dependence.

Problem 2. For the function $\mathbf{v} = \hat{\mathbf{r}}/r^2$, show that,

$$\int (\nabla \cdot \mathbf{v}) d\tau = 4\pi.$$

Approach 1: Use the Gauss divergence theorem.

Approach 2: Use “regularization”. Assume $\mathbf{v} = \hat{\mathbf{r}}/(r^2 + a^2)$, with a being a small length, and take $a \rightarrow 0$ in the end.